CAPSTONE · SUPPLY CHAIN ANALYTICS

DataCo Supply Chain Performance Analysis

Shipping mode profitability and cost efficiency

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Capstone Project

Problem statement: shipping mode profitability

How can DataCo analyse transaction revenues, discounts, and fulfillment costs across shipping modes to identify the most profitable logistics strategies and reduce low-margin shipments?

Standard Class **Second Class** **First Class** **Same Day**

Analysis focus: compare revenue, profit margin, discount patterns and delivery performance by mode.

Five research objectives

01 Analyze Overall Sales Performance

02 Evaluate Shipping Mode Performance & Fulfillment Efficiency

03 Analyze Product, Category and Department Profitability

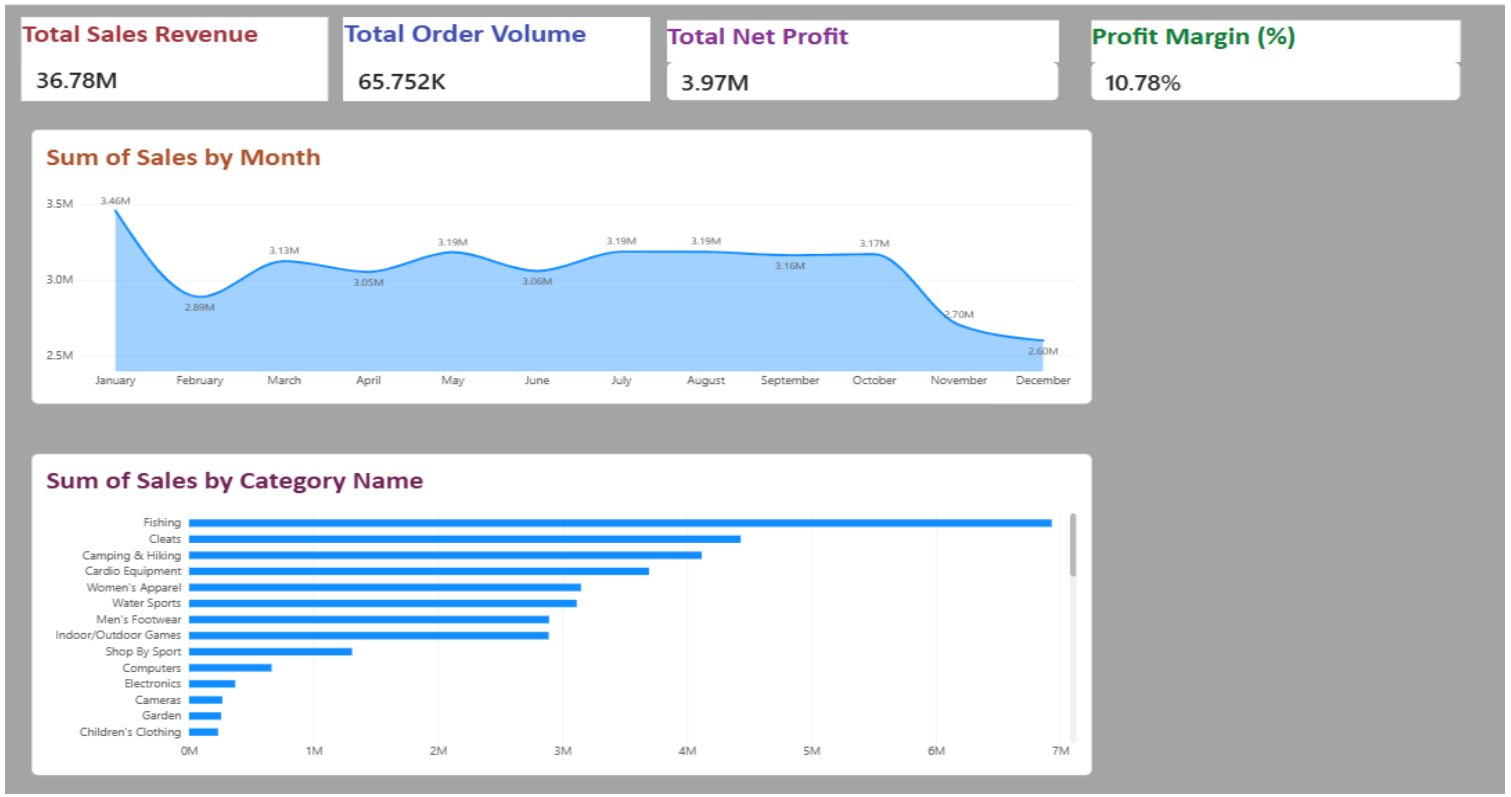
04 Analyze Shipping Mode Profitability & Cost Efficiency

05 Analyze Customer Segment Performance & Purchasing Patterns

Nine connected datasets

Dataset	Records	Main role
Customers	20,652	Customer details and segments
Orders	65,752	Orders and delivery measures
Order Items	180,519	Sales, discounts and profit
Products	118	Product reference
Categories	51	Product categories
Departments	11	Department reference
Shipping	65,752	Shipping mode and dates
Geography	64,990	Markets and locations
Payment Types	4	Payment reference

Objective 1: Analyze Overall Sales Performance



Performance & core insight

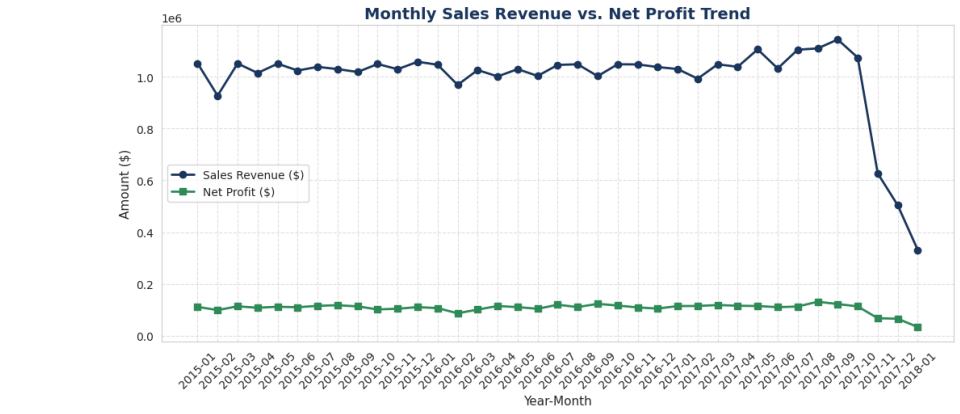
Revenue: USD 36.78M
Unique orders: 65,752
Recorded profit: USD 3.97M
Profit margin: 10.78%

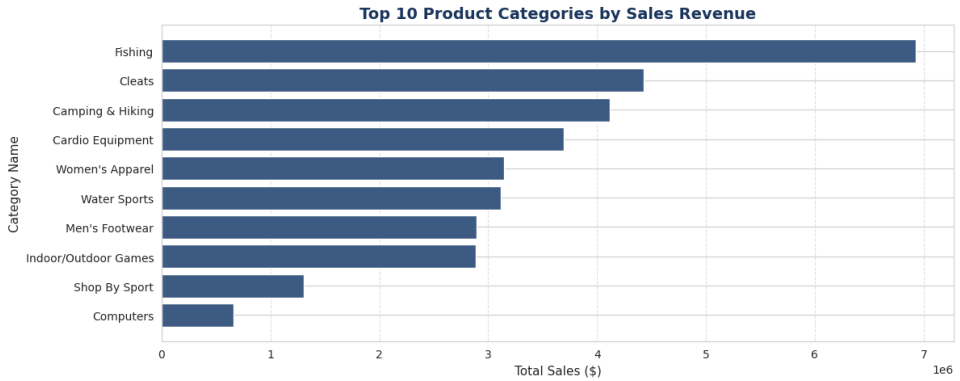
Core insight
About USD 10.78 profit per USD 100 of sales. Additional unrecorded costs could reduce this return.

TAKEAWAY

$36.78M sales 65,752 orders $3.97M reported profit 10.78% margin

Objective 1: Python sales trends and categories





Observation

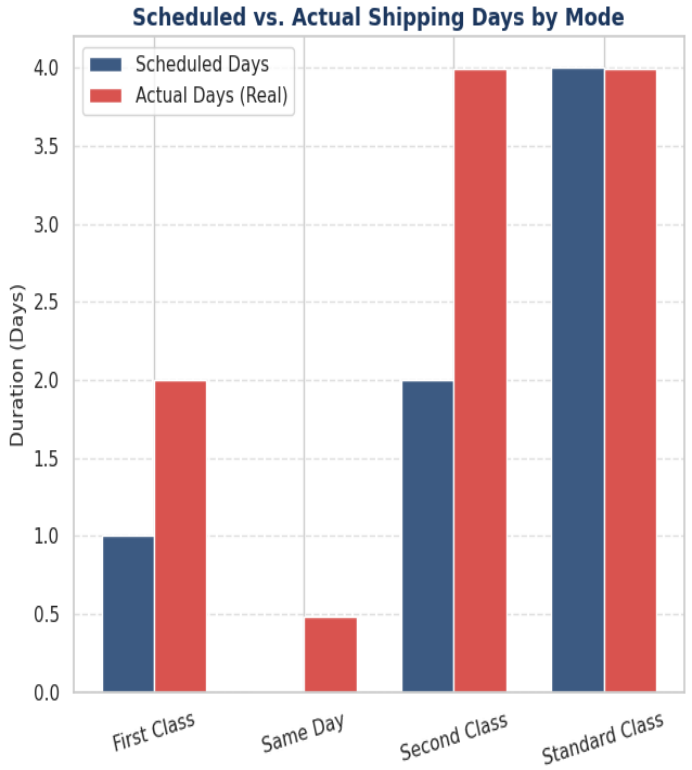
Fishing leads category sales at about $6.9M.

Sales decline at the end of the timeline. Check period coverage first.

TAKEAWAY

Recommendation: protect high-revenue categories and review low-margin items to improve overall profitability.

Objective 2: Shipping Mode Performance & Fulfillment Efficiency





Observation

Standard Class: about 4 days, matching its schedule. Late risk: 38.1%.

Same Day: about 0.5 day on average. Late risk: 45.7%.

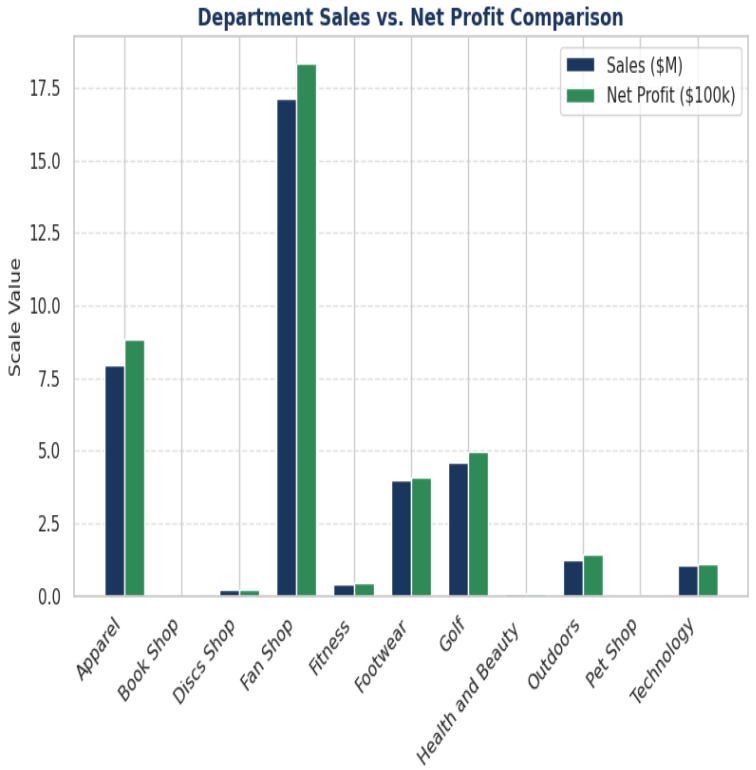
Second Class: about 4 days vs 2 scheduled. Late risk: 76.6%.

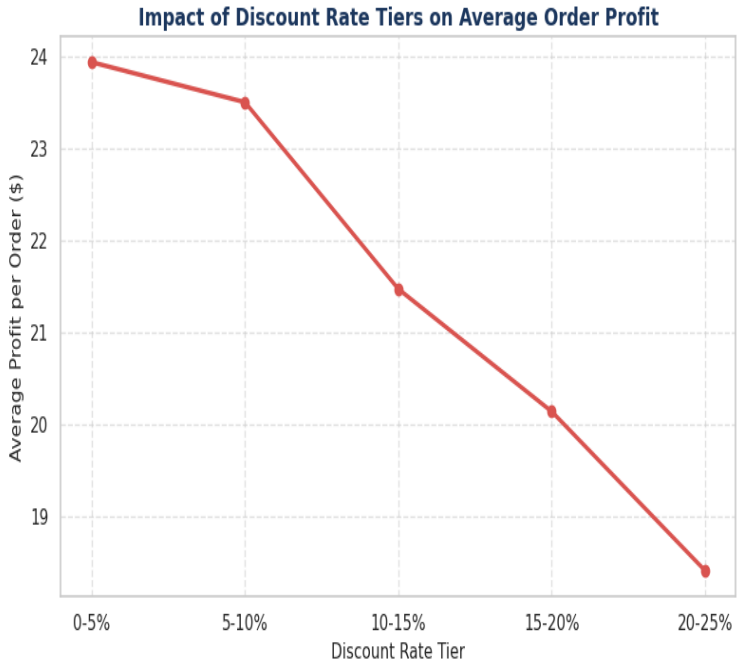
First Class: 2 days vs 1 scheduled. Late risk: 95.3%.

TAKEAWAY

Standard Class best matches its average schedule. Review First and Second Class delivery commitments with carriers.

Objective 3: Product, Category and Department Profitability





Performance & core insight

Fan Shop contributes about USD 17.11M sales and USD 1.83M profit.

Average profit falls from about USD 24 to USD 18.4 as discount bands rise from 0–5% to 20–25%.

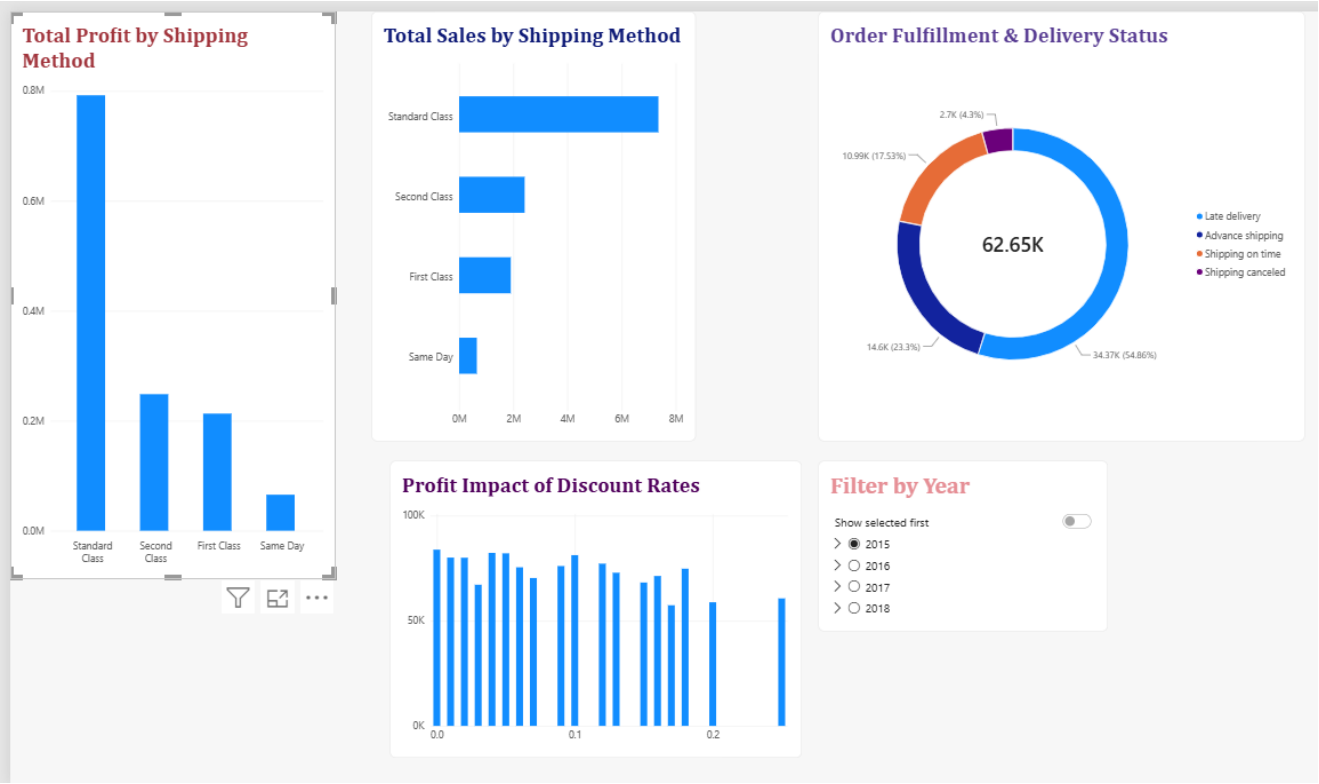
Higher discounts are associated with lower average profit.

TAKEAWAY

Recommendation: review low-margin products and test discount controls by shipping mode. Association does not prove causation.

Objective 4: Analyze Shipping Mode Profitability & Cost Efficiency

Power BI shipping dashboard



Compare revenue and profit margins across shipping modes.

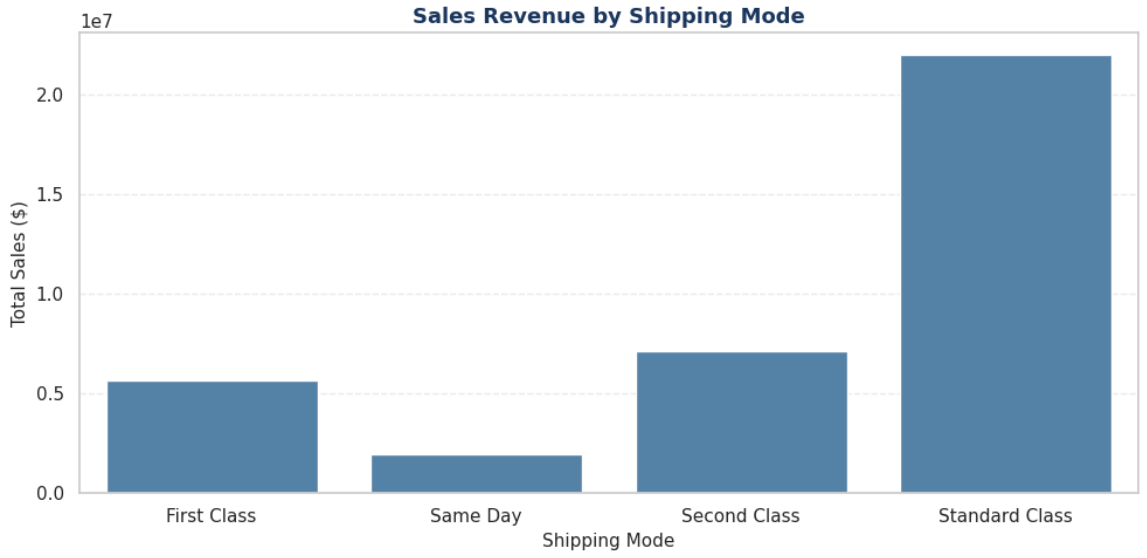
Evaluate operational cost efficiency.

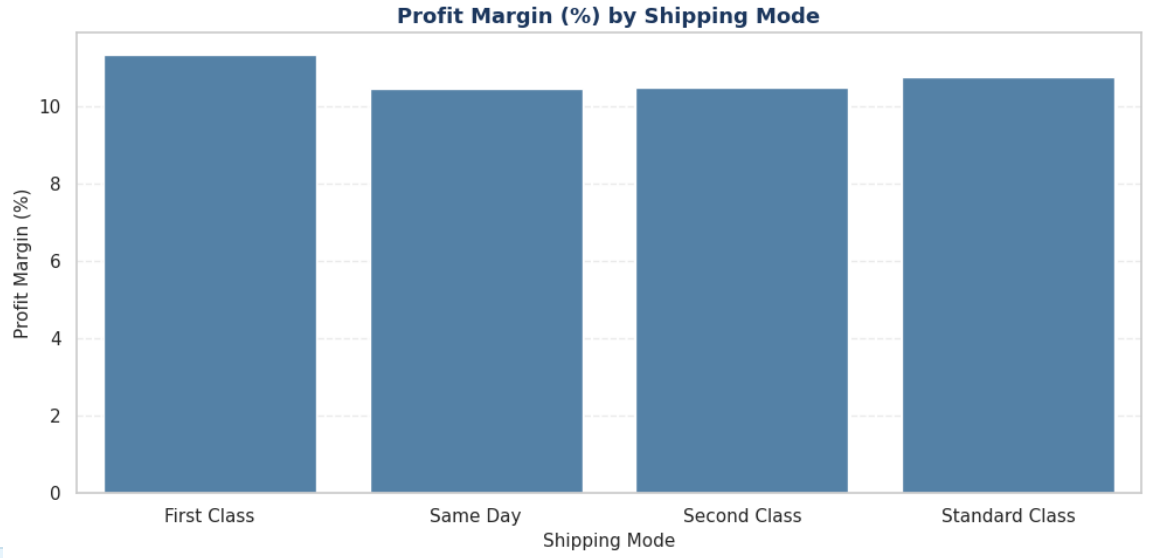
Build a predictive model to classify profitable and unprofitable shipments.

TAKEAWAY

2015 view: late delivery accounts for 54.86% of the displayed delivery statuses.

Objective 4: Python shipping profitability



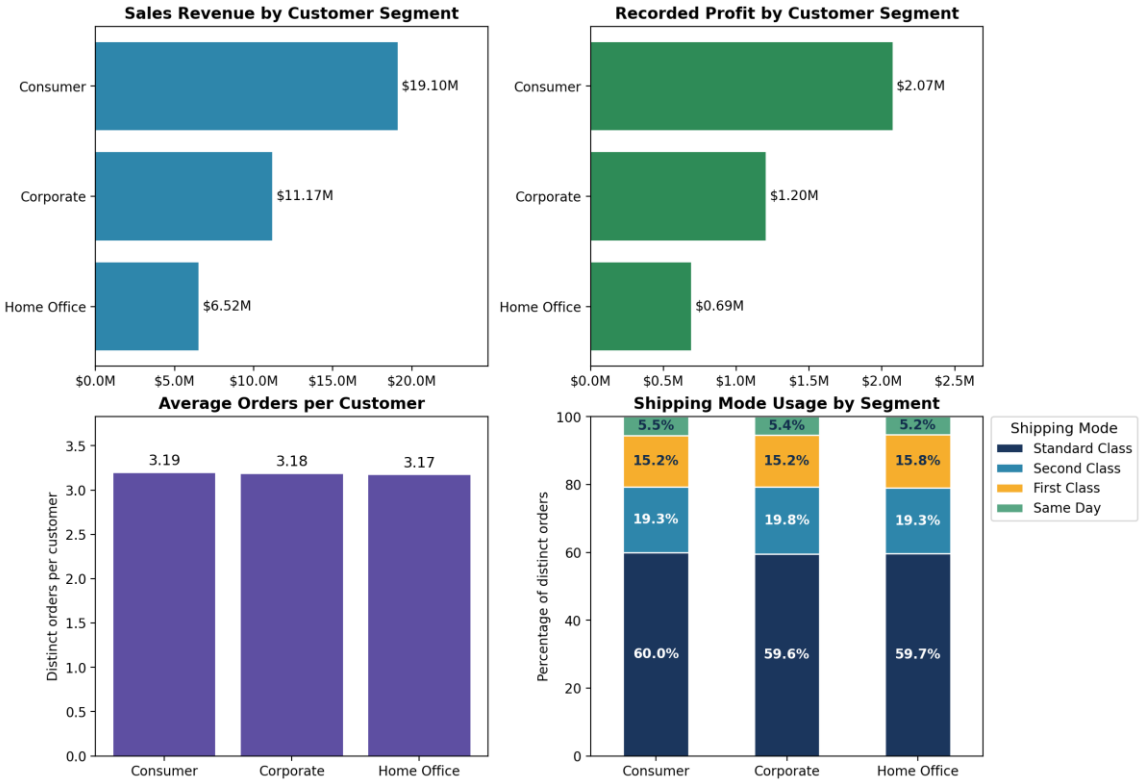


Performance & core insight

Standard Class: USD 22.02M sales and USD 2.37M profit. First Class leads margin at 11.33%, followed by Standard 10.76%, Second 10.50% and Same Day 10.45%. Volume and margin identify different leaders.

TAKEAWAY Recommendation: protect Standard Class capacity, review premium service reliability, and add actual fulfillment costs.

Objective 5: Customer Segment Performance & Purchasing Patterns



Observation

1. Sales contribution
Consumer: USD 19.10M (52%). Corporate: USD 11.17M. Home Office: USD 6.52M.

2. Profit contribution
Consumer: USD 2.07M. Corporate: USD 1.20M. Home Office: USD 0.69M.

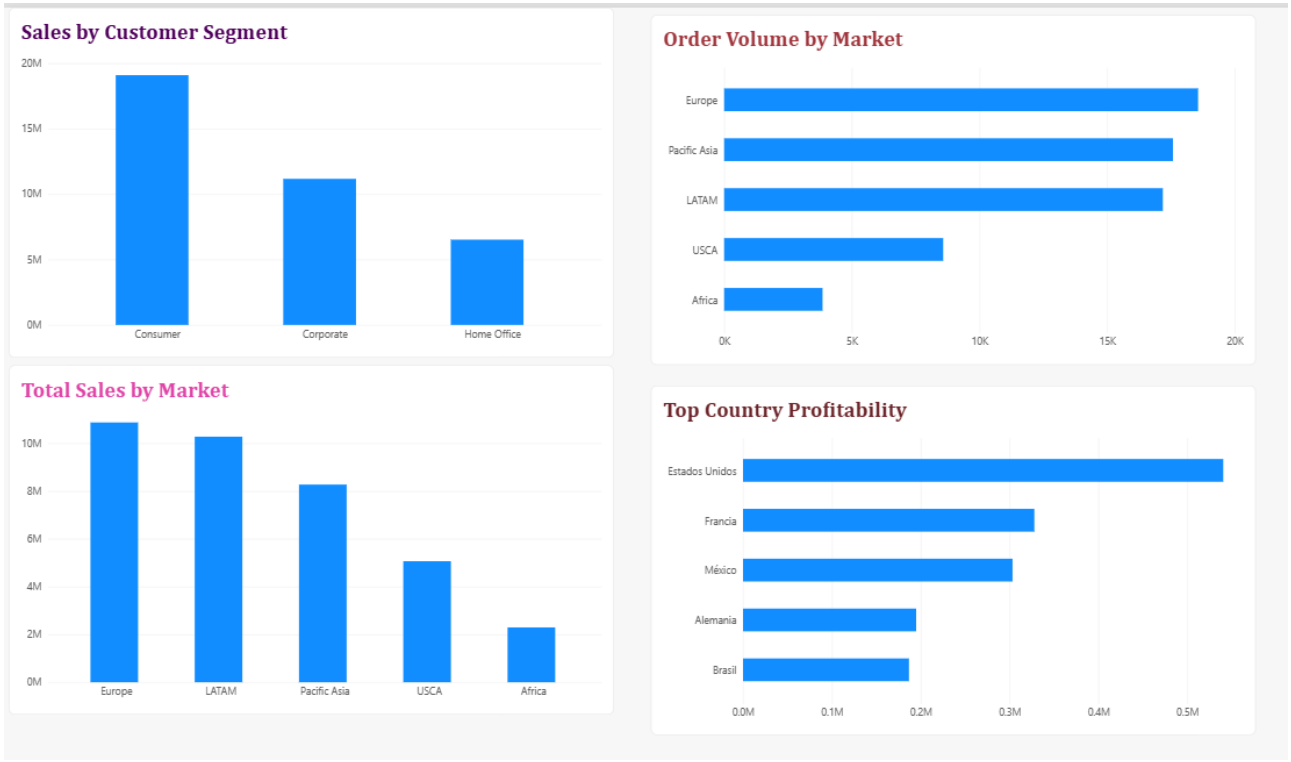
3. Purchasing patterns
Orders per customer: 3.19, 3.18 and 3.17. Frequency is similar across segments.

4. Shipping mode usage
Standard Class represents about 60% of orders in every segment.

TAKEAWAY

Core insight: Consumer drives scale, while ordering and shipping patterns are similar. Recommendation: maintain reliable Standard Class delivery and monitor profit across segments.

Objective 5: Power BI customer dashboard



TAKEAWAY

Consumer leads sales. Europe and LATAM lead market revenue.

Recommendation: support Consumer demand with reliable Standard Class service. Usage does not prove personal preference.

Business recommendations

Evidence	Recommended action
First and Second Class miss average schedules	Review service promises and delay patterns
Average profit falls across discount bands	Pilot discount approval thresholds
Standard Class leads total profit	Protect capacity and monitor its service level
First Class leads margin but misses its schedule	Balance margin against premium service reliability
Actual fulfillment cost data is unavailable	Add carrier charges and handling costs by mode

Conclusion and next steps

Sales and profit: DataCo generated USD 36.78M in sales and USD 3.97M in recorded profit, with a 10.78% margin.

Shipping strategy: Standard Class contributes the most total profit (USD 2.37M). First Class has the highest margin (11.33%), but averages 2 delivery days against 1 scheduled. Review First and Second Class delivery commitments.

Discounts: higher discount bands are associated with lower average profit. Review discounts and low-margin products.

Customers: Consumer contributes about 52% of sales. Standard Class accounts for about 60% of orders in every segment.

Main conclusion: choose shipping strategies using profit, delivery reliability and actual fulfillment costs together.

Thank you · Questions and discussion